Iheb Lamouchi

(+216) 24 629 102 | iheblamouchi.ik@gmail.com | linkedin.com/in/iheb-lamouchi | github.com/iheblam | Tunisia

Third-year Data Science & AI Engineering student at TEK-UP, building RAG pipelines, ML systems and full-stack apps.
Seeking an end-of-study (PFE) internship in AI / Data or full-stack engineering from January 2027

EDUCATION

TEK-UP University 2024 – 2027 (expected)
Engineering Degree — Data Science & AI *Tunisia*

ISTIC, Borj Cedria 2021 – 2024
Licence in Computer Science *Tunisia*

Baccalaureate — Computer Science 2021

WORK EXPERIENCE

Software Engineering Intern Jul – Aug 2026
BNA — Banque Nationale Agricole *Tunisia*

- Built an offline RAG legal assistant over 4 Tunisian legal codes (∼1,700 articles) for the bank's legal department, using hybrid retrieval (dense + BM25 with reranking) and a local Qwen 2.5 3B via Ollama — fully offline, so sensitive data stays on-premise
- Added a citation guard that checks every cited article and penalty figure against the retrieved sources and refuses rather than hallucinates, reaching 84% end-to-end accuracy and a 95.6% retrieval hit rate on a self-built 50-question ground-truth benchmark
- Built the case-management platform the assistant lives in (Spring Boot, Angular, MySQL): cases, procedures, fees, a proposal/approval workflow with a full audit trail, and JWT RBAC validated in both Spring and FastAPI

Web Development & DevOps Intern Feb – May 2024
Sagemcom *Tunisia*

- Set up a full CI/CD pipeline (Jenkins + Docker + SonarQube) that automated deployment and enforced code quality gates, eliminating manual release steps
- Built a project management app (Angular + Django) with JWT-based role access control, containerized with Docker for consistent dev-to-staging promotion

MEAN Stack Development Intern Aug – Sep 2023
Hammamet Informatique Services *Tunisia*

- Delivered a full-stack e-sport platform (MongoDB, Express.js, Angular, Node.js) with JWT-secured routes, team profiles and live match scores

PROJECTS

Learnova — AI-Powered Adaptive Learning Platform

- **Stack:** FastAPI · Next.js · PostgreSQL · Qwen 2.5 7B (Ollama) · multilingual-e5-base · BAAI/bge-reranker-base · PyMuPDF · Surya Layout AI · Tesseract OCR
- Full-stack platform that turns any course PDF into a personalized study space: auto-generates flashcards, chapter summaries and a difficulty-calibrated question bank, then re-targets each new exam toward the concepts the student keeps getting wrong
- Built the ingestion pipeline end-to-end — OCR, layout detection, heading-aware chunking (∼320 tokens/chunk) and 768-dim vector embeddings stored directly in PostgreSQL; concept deduplication via greedy cosine merge reached 92% precision
- Built a two-stage short-answer grader (bi-encoder cosine for recall, cross-encoder reranker for precision-sensitive cases); an adaptive retry loop raised average scores by 18% on re-attempt

TuniTech Advisor — Smart Phone Recommendation System

- **Stack:** FastAPI · React 18 + TypeScript · MongoDB Atlas · Scikit-learn · MLflow · Pandas · Docker · GitHub Actions · Vercel · Render
- Unified 1,236 phones across 5 Tunisian platforms with inconsistent schemas — normalized specs, resolved duplicates, handled missing values — then benchmarked KNN, RF and XGBoost with full MLflow tracking
- Deployed end-to-end: Dockerized FastAPI backend on Render, React frontend on Vercel, MongoDB Atlas for persistence; CI/CD via GitHub Actions on every push; weekly auto-scraper (APScheduler) with per-store timeout handling keeps the dataset fresh

Licence Plate Recognition & Vehicle Speed Estimation — Computer Vision Pipeline

- **Stack:** Python · YOLO11 · fast-plate-ocr · EasyOCR (English + Arabic) · SORT · OpenCV · ONNX Runtime
- Built an end-to-end pipeline that detects, reads and tracks plates in images, videos and live streams, supporting Latin and Arabic-script plates with country prediction
- Added tiled detection for small or distant plates, SORT tracking, plate-to-vehicle linking and multi-frame voting to combine inconsistent OCR readings
- Evaluated on the OpenALPR benchmark: **83.6% plate detection mAP@50 across 444 images** and **86% exact plate recognition accuracy on a balanced 150-image sample**
- Implemented vehicle speed estimation using camera calibration and road-plane homography, with checks for occlusion; exported annotated media and Excel/CSV/JSON reports

PC Repair System — Full-Stack AI Support Platform

- **Stack:** Django REST Framework · React 19 + Vite · PostgreSQL · JWT · Tailwind CSS · Groq API · Gemini API · Railway · Vercel
- Built a full-stack PC repair support platform with role-based access (client, technician, admin), an AI troubleshooting chat powered by Groq and Gemini, a support ticket workflow with file attachments and a searchable issue library with helpful votes
- JWT authentication with automated email flows for password reset and technician onboarding; backend deployed on Railway (PostgreSQL), frontend on Vercel

TECHNICAL SKILLS

LLMs & RAG: Ollama, Sentence-Transformers, RAG pipelines, Prompt Engineering, Vector Embeddings, Cross-Encoders, FAISS, pgvector

Machine Learning: Scikit-learn, TensorFlow, YOLOv11, OpenCV, Pandas, NumPy, MLflow

Backend: Python, Java, Spring Boot, FastAPI, Django, Node.js, GraphQL, REST APIs, PostgreSQL, MySQL, MongoDB, Redis

Frontend: Angular, React, Next.js, TypeScript

DevOps & Tools: Docker, Jenkins, GitLab CI, SonarQube, GitHub Actions, Git, IntelliJ, GitHub Copilot CLI

CERTIFICATIONS & LANGUAGES

PCAP — Certified Associate in Python Programming (Python Institute) | Cisco Data Science Essentials | Version Control (Meta)

Languages: Arabic (Native) | French (Fluent) | English (Fluent — IELTS Certified)